

The Variance and Return Premiums across Assets ¹

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Abstract

This paper measures variance risk premiums in option returns on twenty different futures, including the S&P 500, U.S. Treasuries, currencies, energy, metals, and grains. We implement a novel model-free methodology that constructs tradable option portfolios which are highly correlated with realized variance. Almost all assets earn negative variance risk premiums. A negative variance risk premium can cause the [Beason and Schreindorfer \(2022\)](#) dissection puzzle where most of the underlying asset return premium is earned in the left tail of the asset return distribution. When variance is negatively correlated with asset returns, the negative variance risk premium enhances the left tail of the asset return distribution. But when variance is positively correlated with asset returns, the variance risk premium increases returns in the right tail.

Keywords: VIX, variance risk premium, equity premium, commodities, volatility, futures

JEL Classification: G1, G12, G13, G23

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1. Introduction

The negative variance risk premium on option returns is a well-documented phenomenon. Understanding the variance risk premium and its relation to the equity premium is a fundamental issue in asset pricing. Most of the evidence in the literature to date is based on options on U.S. stocks and stock indices. However, this evidence does not indicate whether the properties of the variance risk premium for U.S. equity options are sample-specific or whether they also describe other assets. In addition to having a large negative variance risk premium, the S&P 500 displays strong negative option skewness and has a strong negative correlation between variance and stock returns. We examine the patterns of variance risk premiums and underlying asset return premiums in other option markets including bonds, currencies, and commodities (energy, metals, and grains). This rich cross-section of assets allows us to assess whether the results for U.S. stocks and indices are typical or unusual.

To measure the variance risk premium, we implement a novel model-free methodology that does not require approximation or interpolation of option prices. Instead, it is based on tradable option portfolios. Although other measures of variance are possible, e.g., [Martin \(2017\)](#), our option portfolio returns are particularly good measures of the “price of variance” because they are 99% correlated with realized asset return variances. We find that all assets except the Swiss Franc earned negative variance risk premiums, often significantly negative. In this respect, U.S. stock options are typical. However, U.S. stock returns have the largest negative correlation with variance. All other assets have weaker correlations with variance, and 40% of the assets have positive correlations between futures returns and variance.

In addition to measuring variance risk premiums, we show how the variance risk premium interacts with the return premium.² [Beason and Schreindorfer \(2022\)](#) emphasized that most of the U.S. equity premium comes from negative returns. This finding quantitatively contradicts

²In the following, we use “return premium” to refer to the underlying futures return premium for brevity.

conventional theoretical models which predict that investors also demand a substantial positive return premium for upside risk. We confirm Beason and Schreindorfer's finding for U.S. stocks and for our overall sample of assets. We show both theoretically and empirically that a negative variance risk premium amplifies this effect when variance is negatively correlated with futures returns. In contrast, we also show that futures returns that are positively correlated with variance earn return premiums in both the left and right tails.

Conventional models predict that the variance risk premium is related to systematic risk that is correlated with variance. For example, the [Merton \(1973\)](#) ICAPM associates return premiums to the risk of intertemporal investment opportunities. Merton's model therefore interprets variance as a systematic *production* risk factor. The [Breedon \(1979\)](#) consumption CAPM interprets systematic variance in terms of covariance with *consumption* growth. [Drechsler and Yaron \(2011\)](#) offer a preference-based explanation of the variance risk premium when variance is persistent enough to affect long-run risk. Other theories consider economic explanations besides systematic risk. [Cochrane and Saa-Requejo \(2000\)](#) show that out-of-the-money (OTM) options are difficult to hedge, particularly OTM put options. [Bakshi et al. \(2010\)](#) propose that short-sale constraints create a demand for options. In response, market makers increase the quoted option prices to earn compensation for holding option risk, and the higher prices cause lower option returns. We find that returns to variance risk are not highly correlated across different assets. This suggests that the negative variance is a feature of option option contracts and option markets, not driven by systematic risk.

The literature to date has extensively documented the existence of a variance risk premium in options on U.S. stocks and indices but has used different methodologies. For example, [Fleming \(1998\)](#) shows that the implied volatility of S&P 100 options exceeds the realized volatility. [Bakshi and Kapadia \(2003\)](#) study delta-hedged option portfolios on the S&P 500 and find a negative variance risk premium. [Carr and Wu \(2009\)](#) and [Goyal and Saretto \(2009\)](#) also find a negative variance risk premium on portfolios of options on individual stocks. Other papers confirm negative vari-

ance risk premiums in stocks, commodities and Treasuries. [Choi et al. \(2017\)](#) find that the variance risk premium in Treasuries is negative and economically large. [Prokopczuk and Chardin \(2014\)](#) find negative variance risk premiums in commodity markets, although [Trolle and Schwartz \(2010\)](#) show that negative variance risk premiums in gas and oil have smaller Sharpe ratios than simply shorting the S&P 500. [Dew-Becker et al. \(2017\)](#) study returns of short-term and long-term straddles across commodities, currencies, equities and Treasuries and also find that realized volatility has a negative risk premium. These studies of diverse markets are difficult to compare because they employ different methodologies to measure the variance risk premium. Some papers rely on models to construct option portfolios and to infer the variance risk premium. Other studies use interpolated prices based on implied volatilities, and their results lack interpretation as returns on tradable strategies.

In this paper, we measure the variance risk premium consistently across assets by constructing model-independent option portfolios. This approach is easily implementable in practice and does not rely on interpolation nor on any assumptions about the underlying model. In addition to representing the price of a tradable option portfolio, our methodology overcomes some drawbacks of the Chicago Board of Options Exchange VIX formula ([CBOE \(2019\)](#)). For example, the CBOE VIX formula occasionally produces negative values for implied variance, but our methodology does not. In addition, our model-free returns produce higher correlations with realized variance than the CBOE VIX formula. Our returns are unambiguously good measures of the variance risk premium because they are highly correlated with realized variance, with a median correlation across assets above 99%.

We find that some features of S&P 500 options are typical, while other features do not characterize most assets. All assets exhibit a volatility smile across different strike prices wherein OTM options are more expensive than at-the-money (ATM) ones based on the Black-Scholes formula. However, S&P 500 options have the most pronounced smirk, where OTM put options have higher implied volatilities than corresponding OTM calls. In contrast, many assets, including agricultural

commodities, the Japanese yen and the Swiss franc, have an inverse smirk with higher prices for OTM calls than for OTM puts. The negative (positive) slope of the smirk is associated with negative (positive) correlations between futures returns and option returns across assets as predicted by [Heston \(1993\)](#) stochastic volatility model.

Inspired by the observation that S&P 500 options exhibit different option characteristics than options on other assets, we next measure the variance risk premium of the assets. We find that the large negative variance risk premium of S&P 500 options describes the pre-COVID sample. During the March 2020 COVID month, option returns exceeded *2000% per month*, bringing the average S&P 500 variance risk premium to only -11.9%, or roughly half the pre-COVID level of -22.9%.³ Other assets also experienced large and positive variance returns during COVID, but these effects were much smaller than for the S&P 500. As a result, the S&P 500 has a smaller negative variance risk premium over the whole sample than most other assets. Agricultural commodities and 10-year Treasuries have the largest negative variance risk premiums. While 19 out of 20 assets earned negative variance risk premiums in our sample, the Swiss franc is the only asset with a positive variance risk premium.

Our variance risk premium results help to understand why [Beason and Schreindorfer \(2022\)](#) found that almost all the equity premium of the S&P 500 index is associated with negative returns.⁴ This is a puzzle because standard asset pricing models with monotonic pricing kernels predict a small or negative equity premium for payoffs during market downturns, and a large equity premium for payoffs during market booms. We show that one can rationalize the puzzling finding of [Beason and Schreindorfer \(2022\)](#) by including a negative variance risk premium that is negatively correlated with the underlying asset returns, as is the case for the S&P 500 index. We illustrate that the negative variance risk premium and the negative correlation fatten the left tail of the risk-neutral return distribution so that most of the return premium is earned in the left tail. We also

³[Dew-Becker and Giglio \(2022\)](#) find a similar result from the Covid period.

⁴Beason and Schreindorfer perform a model-free extension of a long literature on fat option tails. See [Schreindorfer \(2019\)](#) for a model-based anatomy and further references.

show that asset returns that are positively correlated with variance earn positive return premiums in their right tails. In other words, these positively correlated futures do not exhibit Beason and Schreindorfer’s dissection puzzle.

The remainder of the paper is organized as follows. Section 2 describes the data and methodology and graphs option curves. Section 3 measures historical variance risk premiums on 20 different futures. Section 4 describes the [Beason and Schreindorfer \(2022\)](#) dissection puzzle and resolves it in terms of the variance risk premium. A final section concludes.

2. Data and Methodology

2.1. Data and Futures Statistics

We study options on twenty different futures. The data come from IVolatility based on options traded on three major exchanges: CME, NYMEX and ICE. We focus on currencies (Australian dollar, British pound, Canadian dollar, Swiss Franc, Japanese yen), equities (S&P 500 Index), fixed income (10-year and 30-year U.S. Treasury bonds) and commodities (crude oil, natural gas, gold, silver, copper, live cattle, soybean meal, soybean oil, soybeans, wheat, coffee, corn). The earliest starting date is January 2006, and the end dates are generally in July-October 2020 (see the last three columns of [Table 1](#)). We apply standard filters outlined in the methodology of the CBOE VIX calculation (see [CBOE \(2019\)](#)) to filter out unreliable data points. For example, we exclude option prices with zero bids and zero open interest.

To construct option portfolios, we use options with 28 days to expiration.⁵ Given that various underlying commodities have different options and futures expiration dates, this approach ensures that the results are comparable across assets. We use options on the most liquid futures contracts for each asset.

[Fig. 1](#) compares the shape of option curves across assets by graphing average Black-Scholes

⁵Some assets have missing data on options with 28 days to maturity in some months, and then we use 29 or 30 days. The results are similar if we use only 28 days for the whole analysis.

implied volatility curves for each asset. To construct these curves, we fitted quadratic regressions of implied volatility versus log-moneyness ($\log(\text{strike}/\text{futures})$) scaled by the average at-the-money volatility. Although the level of implied volatility varies over time, these curves faithfully represent the general shape of the implied volatility dependence on strike prices. The curves are all convex and display a “smile” or “smirk”. This suggests that there might be a positive price premium and negative return premium for options that pay off in the tails.

The curves in [Fig. 1](#) do not distinguish whether the smiles come from fat-tail physical distributions or whether they reflect a risk premium. While the curvature of these option smiles is uniformly positive, there is substantial variation in the slope of the smirks of different assets. The S&P 500 displays the strongest negative slope, followed by crude oil. By contrast, coffee and the Japanese yen have the most positive slopes. We shall show that these smirks are related to the correlation of asset returns with variance, as predicted by the [Heston \(1993\)](#) stochastic volatility model. Section 4 ultimately shows how this correlation effect carries the variance risk premium into the left- or right-tail of the asset return distribution.

[Table 1](#) shows summary statistics on futures returns across the twenty assets. From a US dollar perspective, the Australian dollar futures appreciated, while all other currency futures (British Pound, Canadian dollar, Swiss Franc, and Japanese Yen) lost money during our sample period. Natural gas and wheat lost more than 2% per month, while the S&P 500 gained 0.84%. The risk of the assets also varies significantly. Oil and natural gas futures have the highest risks, with standard deviations of 11% and 11.4%, respectively, while 10-year Treasury note futures have the lowest risk, with 1.4% standard deviation. S&P 500 futures have the most prominent skewness at -1.16 and also a high kurtosis of 12.4. Oil has the highest positive skewness, and the largest kurtosis of 14.5. One must be cautious about exact comparison across commodities because they have somewhat different sample periods. However, the table illustrates that futures exhibit diverse behavior, and that the S&P 500 is atypical because it has the second largest mean (after silver), the largest negative skewness, and the second-largest kurtosis. The diverse behavior of futures across assets and the

differences with the commonly studied S&P 500 index motivate our curiosity about the behavior of non-S&P-500 assets and their variance risk premiums.

2.2. Variance Portfolio Methodology

To explore the variance risk premium in option returns across assets, we aggregate options of all strike prices to form a representative variance portfolio for each asset. It is important to use a methodology that consistently accommodates assets with different volatilities and different available strike prices. For example, some assets might have low volatility with options trading mostly near-the-money, while other assets might have high volatility with options trading also deep-out-of-the-money. We construct option portfolios comparable across assets in a model-free way. We then document that these option portfolios have payoffs that are highly correlated with realized variance, with a median correlation above 99% across assets. In this sense, these are truly “variance portfolios”. By comparison, [Table A.1](#) in the appendix shows that another commonly used approach to measure variance risk premiums using ATM straddle returns has relatively low correlation with realized daily variance. Theoretically, the effects of discrete strike prices, daily hedging, and jumps could undermine the usefulness of variance, but the high correlations in [Table A.1](#) show that our variance portfolios are indeed excellent descriptions of observed variance.

The dominant benchmarks for option volatility use the CBOE VIX formula for S&P 500 Index options ([CBOE \(2019\)](#)). These benchmarks measure the volatility of Treasury bonds (TYVIX), currency (EYZ), and individual stocks (VXAPL, VXAZN, VXGOG, VXGS, VXIBM). The CBOE formula works well for S&P 500 options with many finely spaced strike prices. But it is not a portfolio, so we cannot compute returns on it. Furthermore, the CBOE formula is inaccurate for some of our futures data and occasionally even yields negative “variance” for the Australian dollar and British pound in our sample. To avoid these limitations, we construct variance portfolios using the more accurate Simpson’s rule of [Heston et al. \(2024\)](#).

Let VIX^2 denote the variance index for horizon T (which equals four weeks in our analysis). Then $e^{-r^f T} VIX^2$ is the price of an option portfolio at time $t = 0$, where r^f is the risk-free interest

rate. The Simpson's formula is:

$$VIX^2 = \frac{2}{T} e^{r^f T} \sum_i \frac{\Delta_i}{K_i^2} \min(Put_i, Call_i) + \epsilon, \quad (1)$$

where

$$\epsilon = \frac{\Delta}{3T} \left(\frac{F - K_0}{K_0^2} + \frac{K_1 - F}{K_1^2} \right) + \frac{e^{r^f T}}{3T} \left(\frac{\Delta - \Delta_0}{K_0^2} Put_0 + \frac{\Delta - \Delta_1}{K_1^2} Call_1 \right) + \frac{1}{T} \left(\log\left(\frac{F^2}{K_0 K_1}\right) - \frac{F}{K_0} - \frac{F}{K_1} + 2 \right),$$

and where Put_i and $Call_i$ represent time $t = 0$ prices of put and call options with strike price K_i and expiration T , K_0 and K_1 denote strike prices just below and just above the forward price F , respectively, and the strike price intervals are $\Delta = K_1 - K_0$ and $\Delta_i = (K_{i+1} - K_{i-1})/2$. Note that this variance index is a portfolio of OTM options, plus a small adjustment between strike prices K_0 and K_1 when the forward price F is not exactly equal to a traded strike price.

Our benchmark portfolio has the advantage of incorporating OTM options of all available strike prices, thereby diversifying any quotation noise in individual option prices. This benchmark maintains consistent sensitivity to variance because it always includes near-the-money options. In this sense, the VIX^2 portfolio is always at-the-money. In contrast, the [Bakshi and Kapadia \(2003\)](#), [Goyal and Saretto \(2009\)](#), and [Dew-Becker et al. \(2017\)](#) approaches of delta-hedging a single option or straddle lose vega sensitivity as the underlying price drifts away from the option strike price.

The prices of our VIX^2 option portfolios measure the variance of options over a 28-day “month” before expiration. When calculating monthly returns, we scale time so that T equals 1 over this period. The corresponding annualized VIX level is $\sqrt{VIX^2 \times 365/28}$. [Table 2](#) shows summary statistics of the VIX level across assets. Energy commodities (natural gas and oil) have the largest and one of the most volatile VIX, whereas currencies and 10-year Treasuries have the lowest and least volatile VIX. The average S&P 500 VIX is ranked 11th across assets but its standard deviation is the third-largest, which illustrates the volatile behavior of equity variance.

In addition to the level of variance measured by our VIX^2 option portfolios, we want to measure delta-hedged returns that are sensitive to variance but not to underlying asset returns. Following [Bondarenko \(2014\)](#) and [Heston et al. \(2024\)](#), the delta-hedge of our ideal VIX^2 portfolio at time t sells $2/F_0 - 2/F_t$ shares of the underlying asset, where F_t denotes the forward price at time t . Importantly, both the value of the VIX^2 index and the hedge are model-free, because they do not require estimation of any parameters. [Carr and Wu \(2009\)](#) show that the resulting hedged portfolio return replicates the realized variance in continuous time, and [Heston et al. \(2023\)](#) derive it as an approximation in discrete-time. We shall empirically demonstrate this with our daily-hedged VIX^2 benchmarks. Specifically, if r_u^F is the daily futures (excess) return on day u , then the hedged VIX^2 return from time t to time T is approximately determined by the realized daily variance of r_u^F

$$r_{t \rightarrow T}^{hedged\ VIX} \approx \frac{RV_{t,T}}{VIX_t^2} - 1, \quad (2)$$

where the realized variance is the sum of squared daily returns over holding period from time t until the option expiration T

$$RV_{t,T} = \sum_{u>t}^{u \leq T} (r_u^F)^2.$$

To measure returns consistently across assets, we construct daily-rebalanced delta-hedged VIX^2 portfolio returns using non-overlapping 28-day periods for every asset.

Our methodology is computationally simple and model-free because we construct a tradeable strategy and do not interpolate the volatility surface, unlike many papers studying the variance risk premium (e.g., [Prokopczuk and Chardin \(2014\)](#), [Choi et al. \(2017\)](#), [Dew-Becker et al. \(2017\)](#)). Strategies that interpolate the volatility surface rely on non-tradable prices in lieu of actual quotes. Our methodology is instead based on observed prices, and is easily implementable in practice. By holding options to expiration, our strategy also avoids paying multiple bid-ask spreads when rolling the option position. This reduces the impact of transaction costs on our results.

3. The Variance Risk Premium Across Assets

This section measures the variance risk premium across all 20 different commodity futures.

3.1. Returns to Variance

The first column of numbers in [Table 3](#) presents the [Carr and Wu \(2009\)](#) variance swap approximation (2) to the average returns on our model-free variance portfolios. This variance swap approximation to a monthly return is just the realized variance over the month divided by the VIX^2 at the beginning of the month, minus 1. The second column presents the actual realized returns on our model-free option portfolios. While these returns often differ by a few percent, the strong correspondence of the numbers across assets supports the usefulness of the approximation.

The Swiss Franc is the only asset with positive average returns on its variance portfolio, whereas all the other assets have negative variance returns.⁶ Because they involve buying OTM options, our model-free variance portfolios have high leverage, and are quite risky as reflected in their standard deviations. The 10-year Treasury notes have the lowest average return to variance portfolios at -25.8% per month followed by corn and live cattle. The S&P 500 has also a large negative average return, but this is highly sensitive to an outlier return during the March 2020 COVID month, when the option portfolio had an extremely large positive return above 2000% per month (see [Fig. A.1](#)). Before this month, the S&P 500 would have had the second largest (negative) variance risk premium of -22.9%. However, with the COVID month, the sample average variance risk premium is only -11.9%, giving the S&P 500 a smaller (negative) variance risk premium than most of the other assets. This shows that historical evidence about the variance risk premium based on S&P 500 returns is sensitive. The other assets also experienced large positive variance returns during the COVID month, but these were orders of magnitude smaller than for the S&P 500, and we only report means over the full sample for these assets.

⁶The Swiss Franc had a large positive outlier return in January 2015 after the peg to the euro was removed.

Fig. 2 shows that the S&P 500 is also atypical in the sense that it has the most negative volatility smile slope, and the largest negative correlation between variance returns and futures returns. The figure also shows that the slope of the volatility smile is increasing in the correlation between variance and futures returns, consistent with the theory of Heston (1993). Intuitively, a negative correlation between variance and asset returns spreads out the left-tail of the return distribution, while a positive correlation spreads out the right tail. The resulting skewness manifests itself in the shape of the option smirks.

3.2. *Correlations of Returns Across Assets*

Table 4, Panel A, reports the correlations of futures returns across assets and shows that these are mostly positive.⁷ The correlation between Australian and Canadian dollar is 72%, and the correlation between gold and silver is 50%. Soybeans, soybean oil and soybean meal, as well as other agricultural commodities like corn and wheat, are also highly correlated. In contrast, U.S. Treasury bonds futures and Japanese Yen are negatively correlated with about half of the other assets. Natural gas futures is negatively correlated with oil and gold futures and with currencies.

Table 4, Panel B, reports correlations of variance returns across assets. Most variance returns are positively correlated with each other, including Treasuries. Unreported diagnostics show that the equally weighted average of variance returns for all assets is highly correlated with the first principal component, and serves a good proxy for systematic variance risk. Oil, S&P 500, and silver variance returns are highly correlated with this average, at 84%, 81%, and 75%, respectively (last row). The correlations of oil, S&P 500, and silver variance returns with each other range from 66% to 75%. Hence, they are all similar proxies for systematic variance risk. The correlations among other variance returns are smaller.

Table 4, Panel C reports the correlation of futures returns with variance returns. Futures returns on a given asset are negatively correlated with variance returns on the same asset in 11 out

⁷Note that one must take care when comparing correlations across commodities when they do not have identical sample periods.

of 20 cases. The most negative correlation is for the S&P 500, at -63%, followed by oil, at -59%. In contrast, the Swiss Franc, Japanese Yen, U.S. Treasury bonds, and natural gas returns are positively correlated with their own variance returns and with the equally weighted average variance return across assets.

3.3. Systematic Risk

We now examine whether the variance risk premium is related to systematic risk. Note that [Table 2](#) and [Table 3](#) show a strong association between the average *level* of VIX, and the standard deviation of hedged VIX^2 returns across assets. The cross-sectional correlation between the average VIX return and its standard deviation is also quite high, at .77 (an R^2 of .60) as illustrated by [Fig. A.2](#) in the Appendix. If this risk is systematic, then it might explain the returns on our VIX^2 portfolios.

In the context of option returns across many different markets, the meaning of “systematic risk” is ambiguous. Some options might inherit systematic risk from their correlation with the underlying futures returns. In the CAPM, systematic risk is measured by covariance with the market portfolio. In more general models, systematic risk might include a pervasive variance factor that affects option returns across assets.

[Table 5](#) explores whether risk in the underlying futures returns can explain returns on the associated VIX^2 option portfolios. Note that our VIX^2 option portfolios are already delta-hedged every day. Nevertheless, returns on delta-hedged options can be correlated with returns on futures due to stochastic volatility. The first column regresses the monthly hedged VIX^2 returns on contemporaneous futures (excess) returns for each commodity.⁸ Under the null hypothesis that futures risk explains options risk, the regression intercept should be zero. The intercepts represent “beta-hedged” returns on our VIX^2 option portfolios. The intercepts for the S&P 500 and Swiss Franc are insignificantly positive. For the rest of the assets, the intercepts are negative, just like

⁸Percentage changes in futures returns prices are excess returns because they represent the profit from an uncollateralized futures position.

the mean VIX^2 returns in Table 1, and many of them are significant. Hence, futures risk does not seem to explain options risk.⁹

If futures risk is idiosyncratic, then it would not explain option returns in terms of a systematic risk premium. Instead, it is possible that option returns have exposure to risk from the aggregate equity market portfolio, or from a systematic variance factor. As mentioned above, unreported diagnostics indicate that the first principal component of the VIX^2 returns across assets is highly correlated with an equally weighted average of these returns. Moreover, the bottom row of Table 4, Panel B shows that the S&P 500 VIX^2 return has 81% correlation with the equally weighted portfolio of all assets' VIX^2 returns. In other words, this systematic variance factor is highly correlated with the S&P 500 VIX^2 return, a proxy for the variance of the equity market portfolio.¹⁰

The right side of Table 5 (columns 6-10) uses the average VIX^2 returns across all assets as a systematic risk factor, by making it the independent variable in regressions of individual asset VIX^2 returns. Again, this risk adjustment does not explain the mean VIX^2 returns for the majority of assets. The intercepts remain largely negative and often statistically significant at conventional levels.

Fig. 3 and Fig. 4 illustrate the failure of futures returns and average VIX^2 betas to explain VIX^2 returns across assets. Fig. 3 shows a positive cross-sectional association between the underlying futures return and the average realized VIX^2 return. Recall that the summary statistics in Table 1 showed 11 out of 20 assets have a positive futures risk premium, whereas Table 3 reports that only 1 out of 20 has a positive variance risk premium. The positive cross-sectional relationship in Fig. 3 means that futures with a larger risk premium tend to have options with a smaller absolute variance risk premium. Similarly, Fig. 4 shows the relationship between average realized VIX^2 returns, and the VIX^2 betas with respect to the average VIX^2 portfolio. Just as in Fig. 3,

⁹Dew-Becker and Giglio (2022) construct beta-hedged option replication strategies with Black-Scholes hedges and reach similar conclusions about the variance risk premium on the S&P 500.

¹⁰The last row of Table 4, Panel B also shows a high correlation of the equally weighted VIX^2 portfolio with oil (84%) and silver (75%).

the relationship is positive. This means that futures with higher systematic covariance risk tend to have a less negative variance risk premium. This finding does not support systematic risk as a promising explanation of the variance risk premium.¹¹

If systematic risk does not explain the variance risk premium, then there should be an alternative economic explanation involving idiosyncratic factors. In a [Black and Scholes \(1973\)](#) world, dealers can perfectly hedge option risk by continuously trading the underlying futures. In practice, short-sale constraints can create a demand for options ([Bakshi et al. \(2010\)](#), [Hong and Stein \(2003\)](#)), leaving option dealers with net short positions. This exposes dealers to volatility risk ([Hull and White \(1987\)](#), [Heston \(1993\)](#)). In addition, OTM options can be particularly difficult for dealers to hedge ([Cochrane and Saa-Requejo \(2000\)](#)). If hedging is difficult or costly, then dealers need more capital to support their option positions. Under such circumstances, it is natural for dealers to increase option prices to compensate for the risk and cost of such positions. According to this explanation, we expect to see larger, more negative variance risk premiums in option markets with illiquidity or fat-tailed price distribution.¹² However, this explanation does not justify why option dealers are able to raise prices. Competitive, well-capitalized and diversified dealers should be able to tolerate idiosyncratic variance risk without a large return premium. Therefore, this explanation entails option dealers with market power, which allows them to earn abnormal profits from selling exposure to variance. Testing such an explanation might require data on dealer inventories to measure the effect of hedging costs on dealer profits and is beyond the scope of this paper.

4. The Variance Risk Premium Explains the Shape of the Equity Premium

The previous sections showed that the variance risk premium is negative for almost all assets, and that option prices reflect the correlation of variance with asset returns. Specifically, a negative correlation between variance and futures returns causes negative option skewness, while a pos-

¹¹It is possible that systematic risk changes over time, but our unreported inspection of subsample returns shows that the shape of option prices and their correlations seem stable in the data.

¹²In earlier version of the paper, we found very mild support for this explanation.

itive correlation causes positive option skewness. It is intuitive that changes in variance that are negatively (or positively) correlated with futures returns can affect skewness by “spreading out” risk in the left-tail (or right-tail) of the futures return distribution. This section shows both theoretically and empirically that the correlation of variance with underlying asset returns does not merely affect the tails of the physical distribution. This correlation also affects where the asset return premium gets earned. Assuming that the variance risk premium is negative, a negative correlation will increase the equity premium in the left tail, amplifying [Beason and Schreindorfer \(2022\)](#)’s equity premium dissection puzzle. But a positive correlation between variance and asset returns is associated with a positive return premium earned in the right tail.

[Beason and Schreindorfer \(2022\)](#) represent the average equity premium (EP) as a weighted average across the entire distribution of returns:

$$EP := E[R_{t+1} - R_t^f] = \int_{-1}^{\infty} R[f(R) - f^*(R)]dR, \quad (3)$$

where $f(R) = E[f_t(R)]$ and $f^*(R) = E[f_t^*(R)]$ are the average payoff and average forward price of an Arrow-Debreu security that pays \$1 if the return on the asset is between R and $R + dR$, and R_t^f is the gross risk-free rate. Differences between $f(R)$ and $f^*(R)$ reflect a local EP in specific Arrow-Debreu return states R . [Beason and Schreindorfer \(2022\)](#) estimate f from realized returns and f^* from option prices using the [Breedon and Litzenberger \(1978\)](#) methodology and normalize the EP across different regions by defining $EP(x)$ as the fraction of the average EP associated with returns below x :

$$EP(x) = \frac{\int_{-1}^x R[f(R) - f^*(R)]dR}{\int_{-1}^{\infty} R[f(R) - f^*(R)]dR}. \quad (4)$$

Increasing segments of $EP(x)$ show states that contribute positively to the EP, while decreasing segments show states that contribute negatively.

While [Beason and Schreindorfer \(2022\)](#) used S&P 500 returns, [Fig. 5](#) reproduces their puzzling finding using both S&P 500 returns and aggregated returns across all of our futures. In both

cases, most of the EP is associated with negative returns, and around 55% of the EP is associated with returns below -10%. For comparison, we also graph the $EP(x)$ curve for the standard [Black and Scholes \(1973\)](#) model. [Beason and Schreindorfer \(2022\)](#) show similar results from a variety of macroeconomic models that nest the Black-Scholes model, including [Barro \(2009\)](#), [Wachter \(2013\)](#), [Bansal and Yaron \(2004\)](#), [Campbell and Cochrane \(1999\)](#). These conventional models predict a monotonically decreasing pricing kernel. This means that investors accept low expected returns to insure market downturns, but demand high expected returns for upside claims that pay off during market booms. Empirically, that is not what we observe in [Fig. 5](#). The empirical curve deviates from theoretical prediction in two important ways. First, it attributes a disproportionate share of the EP to the left tail. More than 70% of the S&P 500's EP and more than 80% of all futures' EP come from negative return states. A second anomaly not emphasized by Beason and Schreindorfer is that the $EP(x)$ function additionally peaks above 100% and then declines slightly in the right tail for the S&P 500. This means that more than 100% of the equity premium is earned from small or negative returns and that investors accept a negative return premium on upside claims.¹³

We can express the [Beason and Schreindorfer \(2022\)](#) dissection (4) in terms of risk-neutral probabilities and relate it to the variance risk premium. The integrand in the numerator of (4) is

$$R[f(R) - f^*(R)]. \quad (5)$$

Empirically, the risk-neutral density $f^*(R)$ has fatter tails than the physical density $f(R)$ ([Bakshi et al. \(2010\)](#)). In other words, the physical density $f(R)$ is less than the risk-neutral density $f^*(R)$ for large absolute returns in the tails of the distribution. For large negative returns R , the integrand (5) multiplies a negative R by a negative difference $[f(R) - f^*(R)]$, contributing a large positive component to the equity premium. For large positive returns, the integrand (5) multiplies a positive R by a negative difference $[f(R) - f^*(R)]$, contributing a negative component to the eq-

¹³This is also consistent with the findings of [Ait-Sahalia and Lo \(1998\)](#).

uity premium. Together, this means that negative returns contribute a large component of the equity premium, and the cumulative equity premium $EP(x)$ could even peak above 100% before declining to 100% in the right tail.

To illustrate the effect of a variance risk premium, consider a simple ARCH model for log-asset returns.

$$\log(R_{t+1}) = r^f + \mu v_t + \sqrt{v_t} z_{t+1} \quad (6)$$

$$v_{t+1} = v_t + \alpha z_{t+1}^2,$$

where r^f is the compound risk-free rate and z_{t+1} follows a standard normal density. Denote the joint probability density of asset returns and variance by $p(R, v)$. We assume that the risk-neutral probability density equals the physical density scaled by an exponential-affine pricing kernel:

$$p^*(R, v) = \frac{p(R, v) e^{\xi v}}{\sqrt{1 - 2\alpha\xi}}.$$

The parameter ξ is the variance aversion, and a positive ξ generates a negative variance risk premium. [Christoffersen et al. \(2013\)](#) show how to algebraically rewrite the physical ARCH process (6) in terms of risk-neutral parameters.

$$\log(R_{t+1}) = r^f - \frac{v_t^*}{2} + \sqrt{v_t^*} z_{t+1}^*, \quad (7)$$

where the risk-neutral variance is

$$v_t^* = \frac{v_t}{1 - 2\alpha\xi},$$

and z_{t+1}^* follows a standard normal density under the risk-neutral measure. When ξ is positive, this

model generates a negative variance risk premium and a risk-neutral variance v_t^* that is higher than the physical variance. In this model, the variance risk premium effectively spreads out the tails of the risk-neutral distribution. When projected onto the futures return, the variance risk premium has a U-shaped effect on the marginal *risk-neutral* density of returns, despite a kernel that depends monotonically on the futures return variance.

Fig. 6 shows the effect of the variance risk premium on the shape of the asset return premium across return states when the average equity premium is 8% per year, the physical volatility is 15% per year and the risk-neutral volatility is 20% per year for the negative variance risk premium case. A negative variance risk premium creates fat tails where most of the asset premium $EP(x)$ (4) is earned from negative returns and the $EP(x)$ peaks above 1 as seen from the red line. A counterfactual positive variance risk premium (with risk-neutral volatility of 10% per year) has the opposite effects as the $EP(x)$ is earned mostly from positive return states and does not peak above 1 as seen from the blue line.

The continuous-time model of Heston (1993) and the asymmetric GARCH model of Christoffersen et al. (2013) also include a parameter that governs the correlation between variance and asset returns. In the presence of a variance risk premium, the correlation between variance and asset returns also affects the shape of the asset premium distribution $EP(x)$. A negative correlation tends to carry the negative variance risk premium into the left tail, while a positive correlation carries it into the right tail.

From this perspective, Beason and Schreindorfer's dissection puzzle can be explained when the underlying asset has a negative variance risk premium and its returns are negatively correlated with variance, as is the case for the S&P 500 index and most of our assets. A negative variance risk premium generates fat tails. A negative correlation between variance and the underlying asset returns generates fatter left tails and as a result, most of the EP is earned in the left tail. In contrast, positive correlation generates fatter right tails and thus less of the EP is earned in the left tail. Fig. 7 illustrates this by separately plotting the EP dissection for assets with positive (above

0.2) and negative (below -0.2) correlation between asset returns and variance. The figure simply equally weights the asset return data within the separate positive and negative correlation categories. Assets with strong negative correlations replicate Beason and Schreindorfer's dissection puzzle. More than 100% of the EP for those assets is earned in negative return states ($R < 0$), whereas positive return states ($R > 0$) contribute negatively to the EP. However, the [Beason and Schreindorfer \(2022\)](#) puzzle does not appear in [Fig. 7](#) for positively correlated assets because for those assets, the positive return states contribute *positively* to the EP and only around 70% of the EP is earned in negative return states. This new finding supports the theoretical interpretation of the EP pattern as a consequence of the variance risk premium.

5. Conclusion

This paper explores the variance risk premium across twenty different assets using a novel methodology that measures returns on model-free option portfolios. We find that some qualities of commonly studied S&P 500 options also describe options in other assets. Specifically, options on all assets exhibit a volatility smile, and all assets but the Swiss Franc have a negative variance risk premium. However, there are substantial differences between S&P 500 options and options on other assets. For example, S&P 500 options have the largest negative option smirk, while many commodity options have a positive smirk. The S&P 500 has the second-largest negative variance risk premium excluding the March 2020 COVID crisis, and experienced the largest positive variance return (above 2000%) across all assets during the COVID crisis.

We find little evidence that the variance risk premium across assets is associated with systematic variance risk. Instead, assets that are positively correlated with systematic variance tend to have smaller (negative) variance risk premiums. Absence of a systematic component to the variance risk premiums supports the [Bakshi et al. \(2010\)](#) explanation that the variance risk premium is caused by short sale constraints and difficulties in hedging tail risk.

We also show that the variance risk premium can potentially explain the puzzling result of

[Beason and Schreindorfer \(2022\)](#) that the S&P 500 equity premium is earned from returns in the left tail rather than the right tail of the return distribution. Our analysis of a larger dataset of 20 assets shows that this puzzle appears only for futures returns that are negatively correlated with variance. These assets have a negative option smirk and earn more than 100% of their return premiums in the left tail. Assets that are positively correlated with variance have positive option smirks and earn positive return premiums in the right tail. We explain that Beason and Shreindorfer's puzzle is consistent with the effect of a negative variance risk premium which manifests itself in the correlated tail of the futures return distribution.

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Tables and Figures

Table 1

Summary statistics of futures returns

The Table shows the mean, standard deviation, Sharpe ratio, skewness, kurtosis and number of observations for 28-day futures returns as well as the start and end dates and the exchange on which options are traded. Means and standard deviations are in %, Sharpe ratio is annualized. The sample period is generally from January 2006 to October 2020; some assets have missing data on options with 28 days to maturity in some months. *, **, and *** indicate statistical significance of the t -statistics for the mean return at the 10%, 5%, and 1% levels, respectively.

	Futures returns						Start date	End date	Exchange
	Mean	Std	Sharpe	Skew	Kurt	Obs.			
Australian Dollar	0.17	3.30	0.18	0.07	4.12	168	2007-02	2020-10	CME
British Pound	-0.13	2.59	-0.18	-0.60	3.96	166	2006-01	2020-10	CME
Canadian Dollar	-0.07	2.30	-0.11	0.02	3.47	152	2006-01	2020-09	CME
Swiss Franc	-0.07	2.94	-0.11	-0.03	5.22	140	2007-01	2020-08	CME
Japanese Yen	-0.07	2.57	-0.11	-0.24	3.64	167	2006-01	2020-09	CME
S&P 500	0.84	5.33	0.58**	-1.16	12.36	167	2006-01	2020-06	CME
10-year UST	0.26	1.37	0.69**	0.21	4.11	120	2006-02	2020-04	CBOT
30-year UST	0.35	2.74	0.47	0.07	3.83	127	2006-02	2020-03	CBOT
Brent Crude Oil	0.46	10.97	0.14	0.61	14.49	94	2011-08	2020-09	NYMEX
Natural Gas	-2.61	11.41	-0.83***	0.26	4.34	162	2006-01	2020-10	NYMEX
Gold	0.79	4.52	0.61**	0.15	2.64	160	2006-01	2020-09	COMEX
Silver	1.01	8.68	0.43	0.30	3.22	152	2006-08	2020-09	COMEX
Copper	0.36	6.48	0.18	0.30	3.95	155	2006-01	2020-07	COMEX
Live Cattle	-0.29	4.74	-0.22	-0.30	4.55	127	2006-01	2020-08	CME
Soybean Meal	0.07	7.38	0.04	0.37	3.26	145	2006-02	2020-08	CBOT
Soybean Oil	0.03	5.56	0.04	0.51	3.60	143	2007-01	2020-08	CBOT
Soybeans	0.56	6.59	0.29	-0.50	4.30	144	2006-02	2020-08	CBOT
Wheat	-2.19	8.18	-0.97***	0.05	3.37	131	2006-02	2020-07	CBOT
Coffee C	-0.69	8.24	-0.29	0.42	3.00	110	2007-01	2020-09	ICE
Corn	-1.49	8.02	-0.69**	0.25	5.98	126	2006-02	2020-07	CBOT

Table 2**VIX summary statistics**

The table shows summary statistics on VIX levels. These are computed as $\sqrt{VIX^2 \times 365/28}$, where VIX^2 is the monthly implied variance. “Mean RVol” is the average realized volatility (also annualized). The sample period is from January 2006 to October 2020.

	Mean	Median	Std (%)	Min	Max	Skew	Kurt	Mean RVol
Australian Dollar	12.4	11.2	5.2	6.4	39.6	2.4	10.9	10.6
British Pound	10.3	9.4	4.0	4.9	29.7	2.3	9.9	8.1
Canadian Dollar	9.7	8.7	3.8	4.6	27.5	1.9	8	7.7
Swiss Franc	10.2	9.8	3.6	3.9	24.2	1.2	5.1	9.4
Japanese Yen	10.9	10.2	3.8	4.9	30.1	1.6	7.8	8.1
S&P 500	22.2	17.6	12.8	9.7	78.0	1.8	6.2	17.3
10-year UST	6.3	5.8	2.0	3.4	14.7	1.3	5.1	4.2
30-year UST	11.1	10.5	3.4	6.5	24.0	1.3	4.9	8.2
Brent Crude Oil	38.5	31.0	24.8	17.3	155.0	3.0	13.1	30.1
Natural Gas	50.6	45.7	17.8	21.9	111.7	1.0	3.7	37.7
Gold	19.0	18.0	6.5	9.5	51.2	1.6	7.3	14.5
Silver	33.7	32.7	12.4	15.6	78.6	1.2	4.7	26.4
Copper	29.4	26.9	11.3	13.5	76.6	1.6	6.3	21.2
Live Cattle	19.1	18.2	7.2	8.1	68.3	3.0	19.6	13.7
Soybean Meal	29.3	28.4	8.2	14.4	53.5	0.7	3.4	20.9
Soybean Oil	24.7	23.3	6.8	15.2	70.5	2.7	16.8	18.1
Soybeans	25.2	24.1	8.1	12.1	56.1	1.4	5.0	18.6
Wheat	33.5	30.4	10.8	19.1	71.9	1.2	4.2	25.1
Coffee C	35.6	34.2	7.4	22.0	56.5	0.4	2.6	25.1
Corn	32.4	31.3	9.3	17.2	64.9	0.7	3.3	22.9

Table 3**Summary statistics of hedged VIX^2 returns**

The table shows the mean variance swap return (VSR) as well as the mean, standard deviation, Sharpe ratio, skewness, kurtosis for hedged VIX^2 returns, the ratio of trading volume, put-call deviations and correlations. Hedged VIX^2 returns are the returns on a dynamically hedged VIX^2 portfolios formed 28 days before options' maturity. Means and standard deviations are in %, Sharpe ratio is annualized. "Mean VSR" is the Carr and Wu (2009) average variance swap return approximation, "Volume ratio" is log ratio of the average futures' trading volume to that of options. "Put-call" is the average absolute put-call-parity deviation computed as the difference between call price and the price of the replicating portfolio (put plus spot price minus the present value of the strike price) as percentage of the spot price across all options and days. $\rho(r_T^F, r_T^{VIX})$ is the correlation coefficient between futures returns and hedged VIX^2 returns. The sample period is from January 2006 to October 2020; some assets have missing data on options with 28 days to maturity in some months. *, **, and *** indicate statistical significance of the t -statistics for the mean VIX^2 return at the 10%, 5%, and 1% levels, respectively.

	Hedged VIX^2 returns and option measures								
	Mean VSR (%)	Mean (%)	Std (%)	Sharpe	Skew	Kurt	Volume ratio	Put-call (%)	$\rho(r_T^F, r_T^{VIX})$
Australian Dollar	-6.4	-2.1	57.4	-0.14	4.0	27.4	10.4	0.08	-0.36
British Pound	-11.8	-9.4	75.2	-0.46	8.44	94.31	10.7	0.08	-0.22
Canadian Dollar	-13.5	-8.8	61.2	-0.52*	4.6	35.0	10.6	0.05	-0.20
Swiss Franc	14.2	14.8	274.3	0.19	11.2	130.5	12.7	0.08	0.30
Japanese Yen	-19.6	-16.6	61.4	-0.98***	2.7	13.1	5.7	0.08	0.27
S&P 500	-13.1	-11.9	192.0	-0.23	8.6	90.7	14.5	0.05	-0.63
S&P 500 pre-Covid	-24.4	-22.9	100.0	-0.86***	4.56	26.01	14.6	0.05	-0.65
10-year UST	-31.9	-25.8	31.5	-2.97***	1.4	6.2	12.7	0.09	0.01
30-year UST	-18.8	-14.4	37.4	-1.41***	1.6	5.9	13.4	0.09	0.16
Brent Crude Oil	-17.2	-12.2	96.2	-0.46	5.4	39.7	16.3	0.05	-0.59
Natural Gas	-15.3	-14.1	62.2	-0.80***	6.4	61.3	10.7	0.09	0.18
Gold	-12.7	-11.7	101.0	-0.42	8.4	88.5	15.9	0.08	-0.08
Silver	-3.7	-5.5	144.1	-0.11	3.5	48.0	12.9	0.33	-0.25
Copper	-18.8	-17.6	45.5	-1.41***	1.8	6.8	14.0	0.12	-0.28
Live Cattle	-25.2	-22.0	52.5	-1.52***	2.8	14.0	13.4	0.12	-0.38
Soybean Meal	-23.2	-15.7	40.9	-1.40***	1.5	5.8	16.6	0.11	0.16
Soybean Oil	-19.2	-16.7	43.3	-1.40***	1.6	7.1	14.0	0.09	-0.04
Soybeans	-24.5	-21.4	42.4	-1.95***	1.5	6.0	15.8	0.08	-0.19
Wheat	-15.2	-13.4	39.6	-1.30***	1.1	4.6	15.7	0.08	0.23
Coffee C	-23.3	-18.5	42.7	-1.57***	1.7	7.9	13.3	0.08	0.38
Corn	-22.8	-22.9	50.3	-1.71***	1.0	6.6	15.4	0.09	0.19

Table 4**Correlations of futures returns and hedged VIX^2 returns**

The table shows monthly correlations of futures returns (Panel A), hedged VIX^2 returns (Panel B), hedged VIX^2 returns and futures returns (Panel C). The last line “Eq. Weighted” shows the correlations of the equally weighted index with the particular asset. In Panel C, “F” stands for futures return, “V” for VIX^2 return.

Panel A: Futures correlations																				
	Australian Dollar	British Pound	Canadian Dollar	Swiss Franc	Japanese Yen	S&P 500	10-year UST	30-year UST	Brent Oil	Natural Gas	Gold	Silver	Copper	Live Cattle	Soybean Meal	Soybean Oil	Soybeans	Wheat	Coffee C	Corn
British Pound	0.45																			
Canadian Dollar	0.72	0.44																		
Swiss Franc	0.39	0.53	0.35																	
Japanese Yen	0.05	-0.02	-0.07	0.30																
S&P 500	0.07	-0.06	0.09	-0.13	-0.20															
10-year UST	-0.01	-0.15	-0.25	-0.12	0.16	-0.47														
30-year UST	-0.10	-0.21	-0.30	-0.32	0.10	-0.26	0.94													
Brent Crude Oil	0.23	0.24	0.27	-0.03	-0.14	0.27	-0.17	-0.23												
Natural Gas	-0.10	-0.01	-0.10	-0.01	-0.11	0.15	-0.10	-0.04	-0.17											
Gold	0.03	-0.06	0.04	-0.03	0.20	-0.12	0.26	0.18	0.16	-0.08										
Silver	0.24	0.11	0.13	-0.01	0.11	0.07	0.10	0.05	0.20	0.10	0.50									
Copper	0.20	0.19	0.29	0.29	-0.10	0.31	-0.34	-0.42	0.31	0.07	0.09	0.24								
Live Cattle	0.17	0.18	0.08	-0.09	-0.22	-0.01	0.02	0.03	0.26	0.04	-0.01	0.03	-0.07							
Soybean Meal	0.20	0.27	0.20	0.32	0.00	0.10	-0.05	-0.15	0.08	0.19	-0.07	0.04	0.24	-0.04						
Soybean Oil	0.27	0.22	0.27	0.25	0.09	0.11	-0.09	-0.19	0.21	0.05	0.06	0.12	0.25	0.05	0.39					
Soybeans	0.22	0.22	0.19	0.21	-0.07	0.28	-0.12	-0.14	0.09	0.24	-0.12	0.08	0.14	-0.02	0.74	0.6				
Wheat	-0.12	0.00	-0.18	-0.10	0.04	0.03	0.09	0.08	-0.10	0.08	-0.06	0.08	0.06	0.10	0.39	0.14	0.43			
Coffee C	0.17	-0.01	0.22	-0.14	-0.12	0.08	0.03	0.02	0.01	0.25	0.26	0.16	0.05	0.10	0.07	0.22	0.26	0.22		
Corn	-0.01	0.08	-0.10	0.01	0.04	0.15	0.09	0.10	0.01	0.33	-0.05	0.12	-0.04	0.11	0.52	0.34	0.68	0.66	0.23	
Eq. Weighted	0.37	0.33	0.31	0.23	0.02	0.36	-0.01	-0.08	0.51	0.48	0.18	0.49	0.39	0.24	0.66	0.57	0.74	0.52	0.43	0.73

Panel B: Hedged VIX^2 returns correlations																				
	Australian Dollar	British Pound	Canadian Dollar	Swiss Franc	Japanese Yen	S&P 500	10-year UST	30-year UST	Brent Oil	Natural Gas	Gold	Silver	Copper	Live Cattle	Soybean Meal	Soybean Oil	Soybeans	Wheat	Coffee C	Corn
British Pound	0.64																			
Canadian Dollar	0.77	0.68																		
Swiss Franc	0.13	0.44	0.27																	
Japanese Yen	0.32	0.17	0.29	0.03																
S&P 500	0.15	0.10	0.10	0.03	0.47															
10-year UST	0.08	-0.03	0.07	0.00	0.23	0.11														
30-year UST	0.08	-0.06	0.06	-0.04	0.28	0.33	0.81													
Brent Crude Oil	0.06	0.14	0.01	0.23	0.52	0.75	0.19	0.18												
Natural Gas	0.01	0.09	0.00	-0.05	0.10	0.08	-0.07	-0.09	0.25											
Gold	-0.01	0.06	0.02	0.00	0.20	0.26	0.02	0.17	0.11	0.03										
Silver	0.00	0.06	-0.05	0.04	0.34	0.66	0.08	0.12	0.66	0.11	0.53									
Copper	0.11	-0.06	0.06	-0.06	0.26	0.33	0.06	0.18	0.27	0.04	0.27	0.37								
Live Cattle	0.35	0.41	0.35	0.02	0.27	0.22	-0.15	-0.07	0.47	0.16	0.05	0.24	0.10							
Soybean Meal	-0.11	-0.06	-0.04	0.00	0.05	-0.01	-0.04	-0.05	-0.05	0.02	0.01	0.09	0.13	0.04						
Soybean Oil	0.01	0.00	-0.07	0.08	0.09	0.13	0.16	0.23	0.11	-0.04	0.24	0.11	0.10	-0.02	0.33					
Soybeans	0.06	-0.04	0.06	-0.02	0.13	0.19	0.01	0.06	0.07	-0.02	0.04	0.12	0.21	0.13	0.57	0.25				
Wheat	-0.06	-0.02	0.00	0.06	0.08	0.08	0.05	0.08	0.10	-0.09	0.13	0.16	0.19	-0.02	0.19	0.10	0.33			
Coffee C	0.18	0.14	0.16	-0.01	0.14	0.03	0.15	0.15	0.26	0.19	0.08	0.13	-0.09	0.27	0.11	0.14	0.21	0.20		
Corn	-0.07	-0.09	-0.10	-0.01	-0.01	-0.22	0.25	0.12	0.11	-0.12	0.15	0.22	0.23	-0.09	0.28	0.04	0.33	0.47	0.04	
Eq. Weighted	0.32	0.32	0.34	0.37	0.55	0.81	0.25	0.31	0.84	0.17	0.45	0.75	0.39	0.39	0.2	0.31	0.32	0.28	0.28	0.21

Table 4
Correlations of futures returns and hedged VIX^2 returns
(Continued from previous page)

Panel C: Futures-Hedged VIX^2 returns cross-correlations																					
	Australian Dollar, F	British Pound, F	Canadian Dollar, F	Swiss Franc, F	Japanese Yen, F	S&P 500, F	10-year UST, F	30-year UST, F	Brent Oil, F	Natural Gas, F	Gold, F	Silver, F	Copper, F	Live Cattle, F	Soybean Meal, F	Soybean Oil, F	Soybeans, F	Wheat, F	Coffee C, F	Corn, F	Eq. Weighted, F
Australian Dollar, V	-0.36	-0.11	-0.25	-0.04	0.17	0.03	0.11	0.03	-0.25	0.03	0.16	0.03	0.03	-0.21	-0.07	-0.03	-0.13	0.06	0.09	-0.03	-0.06
British Pound, V	-0.25	-0.22	-0.21	-0.12	0.04	0.26	0.03	0.04	-0.3	0.04	0.11	-0.02	0.06	-0.26	-0.12	-0.11	-0.12	-0.06	0.01	-0.12	-0.15
Canadian Dollar, V	-0.29	-0.17	-0.20	-0.09	0.07	0.09	0.23	0.15	-0.18	-0.03	0.12	-0.03	0.01	-0.30	-0.10	-0.11	-0.11	-0.10	0.03	-0.14	-0.16
Swiss Franc, V	-0.11	0.13	-0.17	0.30	0.10	0.03	-0.16	-0.18	-0.28	0.00	-0.16	-0.12	0.10	-0.10	0.05	0.00	0.04	-0.02	-0.13	0.01	-0.05
Japanese Yen, V	-0.17	-0.06	-0.11	0.13	0.27	-0.35	0.26	0.25	-0.37	0.02	0.08	-0.08	-0.16	-0.21	0.09	0.09	0.05	0.14	-0.06	0.12	-0.10
S&P 500, V	-0.06	-0.01	-0.09	0.12	0.21	-0.63	0.30	0.20	-0.49	-0.07	0.11	-0.19	-0.22	-0.22	0.08	-0.06	-0.15	0.04	-0.08	-0.07	-0.36
10-year UST, V	-0.19	-0.05	-0.01	0.09	0.11	-0.20	0.01	0.04	-0.13	-0.01	0.09	-0.03	0.09	-0.08	-0.15	-0.07	-0.08	-0.05	0.08	-0.09	-0.07
30-year UST, V	-0.15	-0.03	-0.07	0.03	0.18	-0.38	0.13	0.16	-0.17	-0.09	0.08	-0.03	-0.10	-0.06	-0.14	-0.12	-0.14	-0.03	0.03	-0.10	-0.17
Brent Crude Oil, V	-0.12	-0.12	-0.14	0.16	0.23	-0.46	0.15	0.16	-0.59	0.07	-0.04	-0.21	-0.29	-0.36	-0.01	-0.01	-0.09	0.09	0.06	0.04	-0.35
Natural Gas, V	0.04	-0.07	-0.04	-0.03	0.01	-0.04	-0.04	-0.02	-0.23	0.20	0.04	0.01	-0.01	-0.19	-0.01	-0.03	-0.04	-0.02	0.03	-0.01	-0.01
Gold, V	-0.08	0.01	0.02	0.01	0.00	-0.21	0.11	0.15	-0.13	0.01	-0.08	-0.18	-0.14	-0.08	0.00	-0.12	-0.06	0.01	-0.08	-0.11	-0.13
Silver, V	-0.12	0.01	-0.06	0.09	0.09	-0.35	0.05	0.07	-0.42	-0.01	-0.22	-0.25	-0.17	-0.26	-0.09	-0.16	-0.16	-0.13	-0.08	-0.10	-0.35
Copper, V	-0.13	-0.08	-0.14	-0.17	0.11	-0.22	0.18	0.19	-0.23	0.12	-0.09	-0.06	-0.28	0.00	-0.07	0.02	-0.01	0.15	-0.06	0.08	-0.09
Live Cattle, V	-0.20	-0.11	-0.14	-0.06	0.08	0.04	-0.08	-0.03	-0.25	0.10	0.08	0.06	-0.11	-0.38	-0.18	-0.01	-0.23	-0.15	0.13	-0.16	-0.2
Soybean Meal, V	0.06	-0.10	-0.05	-0.07	-0.15	-0.03	-0.03	0.01	0.00	0.15	-0.01	0.15	-0.14	0.06	0.16	-0.11	0.09	0.10	0.07	0.07	0.09
Soybean Oil, V	-0.04	-0.05	-0.12	-0.03	0.02	-0.16	-0.03	0.04	-0.24	0.08	-0.01	0.05	-0.08	0.08	-0.06	-0.04	-0.13	-0.12	-0.02	-0.02	-0.08
Soybeans, V	-0.02	-0.15	-0.14	-0.11	-0.04	-0.27	0.26	0.22	-0.13	0.06	0.12	0.16	-0.15	-0.02	-0.03	-0.17	-0.19	0.06	0.03	0.02	-0.01
Wheat, V	-0.02	0.02	-0.05	0.03	0.03	-0.12	0.08	0.09	-0.16	0.02	-0.08	-0.03	-0.12	0.09	0.12	-0.07	0.00	0.23	0.08	0.07	0.05
Coffee C, V	-0.07	-0.07	0.00	0.04	-0.03	-0.11	-0.07	-0.04	-0.19	0.16	0.13	0.04	-0.03	-0.20	0.13	0.00	0.12	0.07	0.38	0.11	0.12
Corn, V	-0.02	-0.07	-0.03	0.02	-0.03	0.08	0.12	0.17	-0.16	0.20	-0.18	0.02	-0.02	0.06	-0.04	-0.22	0.06	0.21	0.21	0.19	0.2
Eq. Weighted, V	-0.21	-0.08	-0.20	0.15	0.20	-0.43	0.13	0.12	-0.60	0.05	-0.07	-0.20	-0.19	-0.32	-0.03	-0.13	-0.12	0.00	-0.02	-0.02	-0.29

Table 5**Regressions of hedged VIX^2 returns**

Hedged VIX^2 returns are the returns on a dynamically hedged VIX^2 portfolios formed 28 days before options' maturity. α and β are the estimates of a regression of hedged VIX^2 returns on the futures excess return: $r_t^{VIX} = \alpha + \beta r_t^F + \epsilon_t$. α_{EW} and β_{EW} are the estimates of a regression of hedged VIX^2 returns on the equally weighted VIX^2 return across assets: $r_t^{VIX} = \alpha_{EW} + \beta_{EW} r_t^{VIX_{EW}} + \epsilon_t$. "se" stands for standard error, "IR" for annualized information ratio (the ratio of α to the standard deviation of ϵ_t). The sample period is from January 2006 to October 2020; some assets have missing data on options with 28 days to maturity in some months. *, **, and *** indicate statistical significance of the t -statistics for α and α_{EW} at the 10%, 5%, and 1% levels, respectively.

	α	α se	β	β se	IR	α_{EW}	α_{EW} se	β_{EW}	β_{EW} se	IR _{EW}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Australian Dollar	-0.01	0.04	-6.25	1.26	-0.07	0.03	0.04	0.41	0.09	0.20
British Pound	-0.10	0.06	-6.49	2.21	-0.49*	-0.02	0.06	0.56	0.13	-0.10
Canadian Dollar	-0.09	0.05	-5.25	2.13	-0.54*	-0.04	0.05	0.45	0.10	-0.25
Swiss Franc	0.17	0.22	28.00	7.58	0.23	0.37	0.22	2.11	0.45	0.52*
Japanese Yen	-0.16	0.05	6.49	1.79	-0.97***	-0.07	0.04	0.77	0.09	-0.49*
S&P 500	0.07	0.12	-22.67	2.18	0.17	0.31	0.09	3.47	0.19	1.00***
10-year UST	-0.26	0.03	0.22	2.11	-2.97***	-0.22	0.03	0.27	0.10	-2.59***
30-year UST	-0.15	0.03	2.21	1.20	-1.46***	-0.10	0.03	0.36	0.10	-1.01***
Brent Crude Oil	-0.10	0.08	-5.14	0.74	-0.46	0.10	0.06	1.54	0.10	0.68*
Natural Gas	-0.11	0.05	1.09	0.42	-0.65**	-0.11	0.05	0.23	0.11	-0.65**
Gold	-0.10	0.08	-1.82	1.77	-0.36	0.10	0.08	1.53	0.24	0.40
Silver	-0.01	0.11	-4.13	1.31	-0.03	0.20	0.08	2.30	0.17	0.75***
Copper	-0.17	0.04	-1.97	0.54	-1.40***	-0.13	0.03	0.39	0.07	-1.12***
Live Cattle	-0.23	0.04	-4.22	0.92	-1.71***	-0.18	0.04	0.42	0.09	-1.34***
Soybean Meal	-0.16	0.03	0.87	0.46	-1.42***	-0.12	0.04	0.27	0.11	-1.08***
Soybean Oil	-0.17	0.04	-0.30	0.66	-1.41***	-0.09	0.04	0.47	0.12	-0.79***
Soybeans	-0.21	0.03	-1.22	0.53	-1.82***	-0.15	0.04	0.45	0.11	-1.34***
Wheat	-0.11	0.03	1.11	0.41	-1.03***	-0.08	0.04	0.36	0.11	-0.76**
Coffee C	-0.17	0.04	1.99	0.46	-1.55***	-0.14	0.04	0.37	0.12	-1.23***
Corn	-0.21	0.04	1.17	0.55	-1.53***	-0.18	0.05	0.33	0.14	-1.31***

Fig. 1. Implied volatility smiles

The figure shows scaled implied volatility (scaled by the at-the-money (ATM) implied volatility σ_{ATM} for each asset) as a function of scaled moneyness $m = \log(K/F)/\sigma^{ATM}$, where F is the futures price, K is the strike price of the option, and σ is volatility. The lines plot the function $\sigma(m)/\sigma^{ATM} = \tilde{\alpha} + \tilde{\beta}m + \tilde{\gamma}m^2$ for each asset, where $\tilde{\alpha}, \tilde{\beta}, \tilde{\gamma}$ are the time-averaged estimates of α, β, γ from a series of cross-sectional (across strikes) regressions $\sigma_t(m_t)/\sigma_t^{ATM} = \alpha + \beta m_t + \gamma m_t^2 + \epsilon_t$ for each asset at each time t .

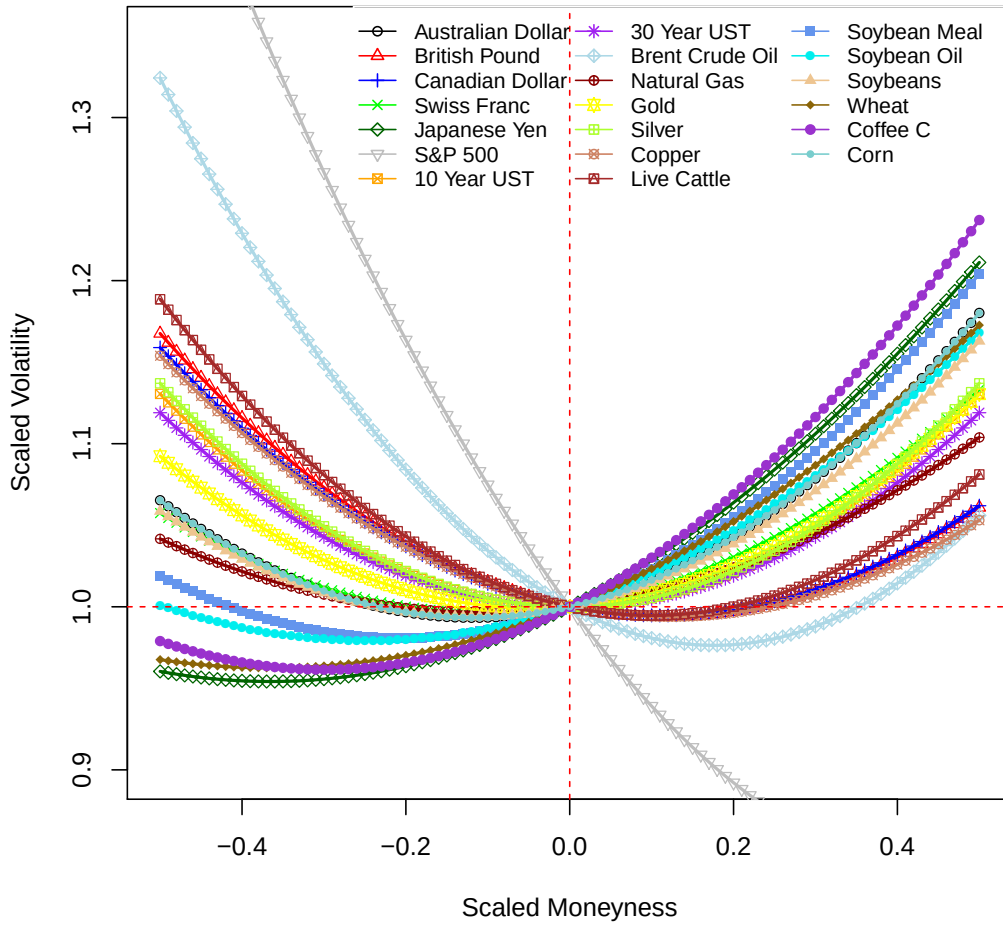


Fig. 2. Variance risk premium and the slope of the volatility smile

The figure shows the correlation between the variance risk premium, measured by the return on the hedged VIX^2 portfolio and the futures risk premium (FRP), measured by the excess return on the futures contract, and the average slope of the implied volatility smile for each asset.

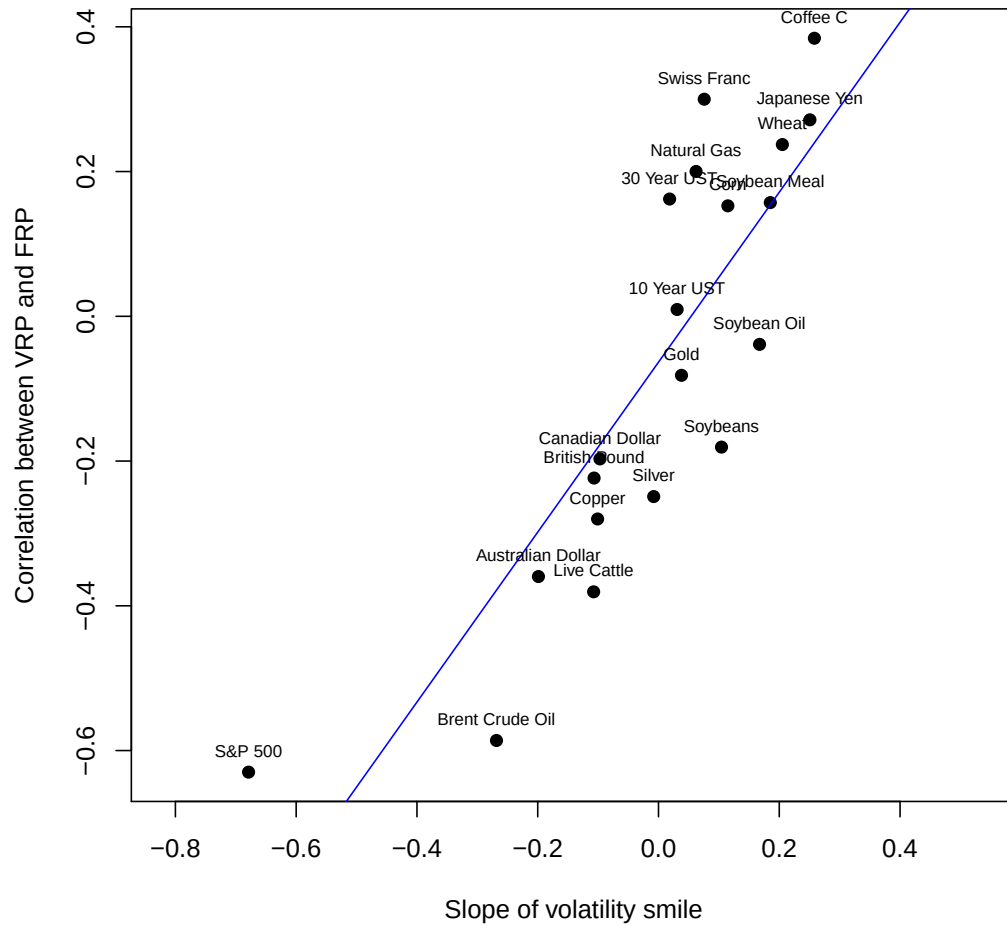


Fig. 3. Variance risk premium and futures risk premium

The figure shows the annualized variance risk premium (measured by the average return on the hedged VIX^2 portfolio) and the futures risk premium (measured by the average excess return on the futures contract).

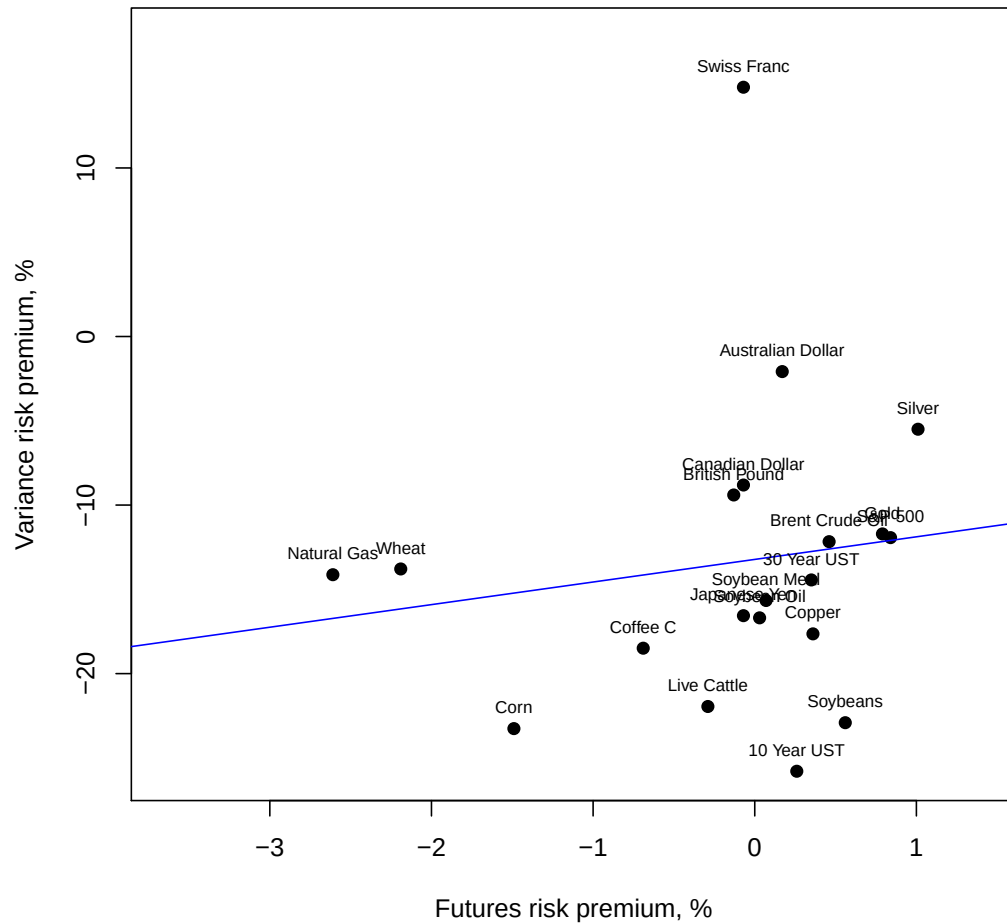


Fig. 4. Variance risk premium and β of systematic variance.

The figure shows the size of the variance risk premium and the β from a regression of individual-asset variance risk premium on the average variance risk premium across assets. The variance risk premium is measured by the average return on the corresponding hedged VIX^2 portfolio.

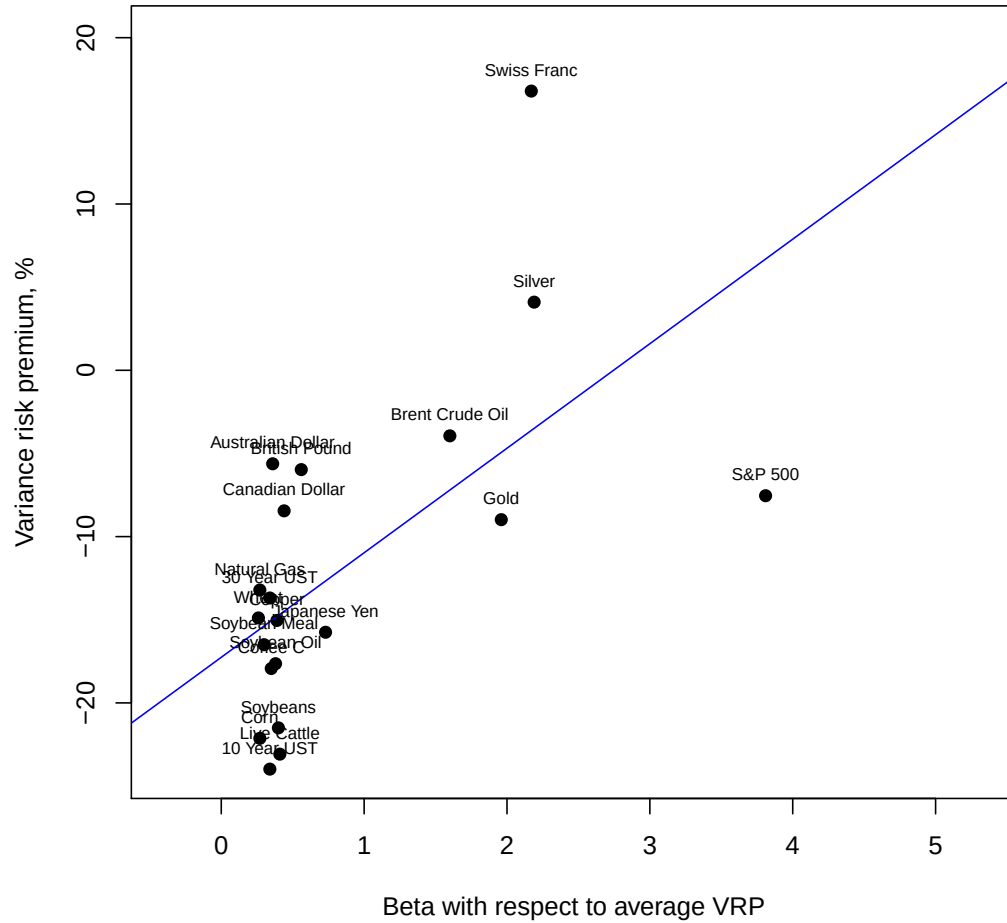


Fig. 5. Sources of the equity premium.

The graph replicates Figure I of [Beason and Schreindorfer \(2022\)](#) for the S&P 500 and for all 20 futures. The lines depict integral (4) as a function of the upper limit of integration, x , and normalized by the total equity premium: $EP(x) = \frac{\int_{-1}^x R[f(R) - f^*(R)]dR}{\int_{-1}^{\infty} R[f(R) - f^*(R)]dR}$. The blue line shows the simple average of all 20 $EP(x)$ curves for the 20 futures.

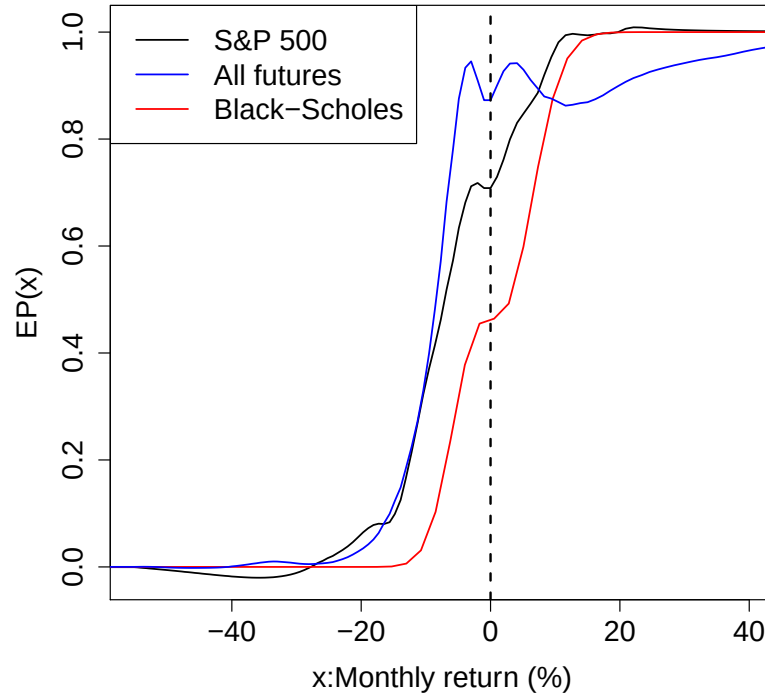


Fig. 6. Sources of the equity premium and the variance risk premium (VRP).

The graph replicates Figure I of [Beason and Schreindorfer \(2022\)](#) for the [Christoffersen et al. \(2013\)](#) ARCH model. Log-returns follow the physical process from Eq. (6) and Eq. (7). The graph shows the effect of the VRP on the shape of the equity premium across return states. The average equity premium is 8% per year, the physical variance is $\nu_t = 15\%^2$ per year, and the risk-neutral variance is $\nu_t^* = 20\%^2$ per year for the negative VRP case, $\nu_t^* = 15\%^2$ per year for the no VRP case, and $\nu_t^* = 10\%^2$ per year for the positive VRP case.

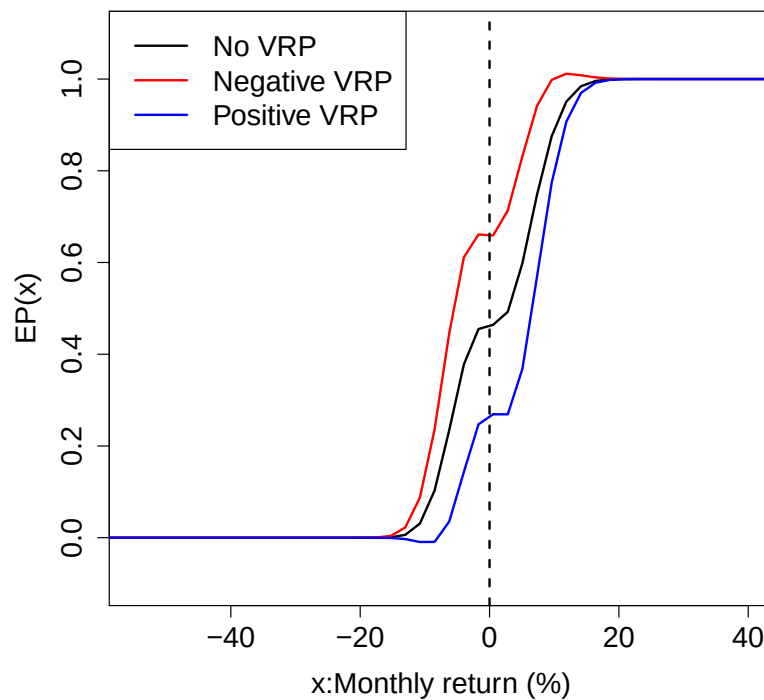
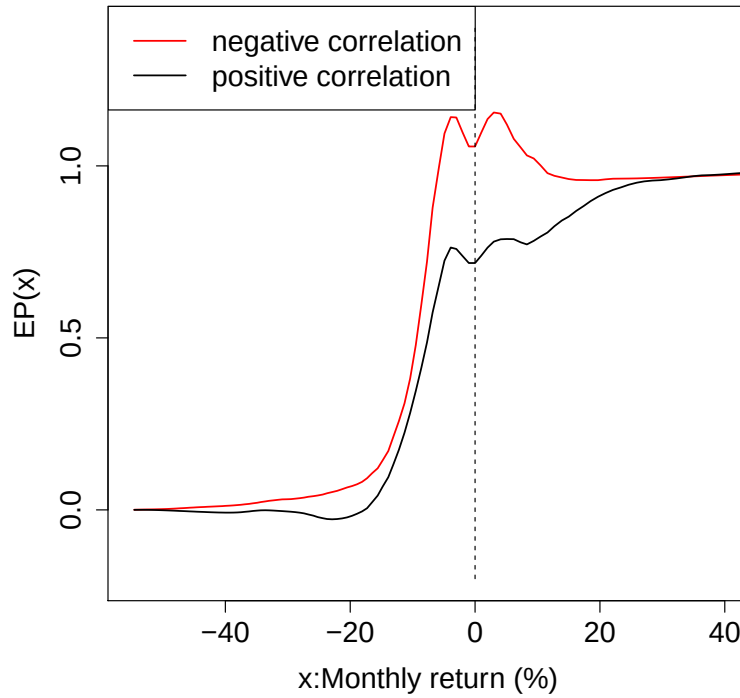


Fig. 7. Sources of the equity premium depend on the correlation between returns and variance.

The graph replicates Figure I of [Beason and Schreindorfer \(2022\)](#) for assets with positive or negative correlation between returns and variance. The assets with positive correlation are Swiss Franc, Japanese Yen, wheat and Coffee C (all have positive correlation above 0.20). The assets with negative correlation are Australian Dollar, British Pound, Canadian Dollar, S&P 500, oil, silver, copper, live cattle (all have negative correlation below -0.20).



Appendix A: Additional tables and figures

Table A.1

Summary statistics comparing hedged VIX^2 returns r^{VIX} , Carr and Wu (2009) approximation of variance swap returns r^{VSR} , and at-the-money straddle returns r^{str} . “SR” stands for Sharpe ratio, ρ for time-series correlation coefficient.

	mean r^{VIX}	SR r^{VIX}	mean r^{VSR}	SR r^{VSR}	mean r^{str}	SR r^{str}	$\rho(r^{VSR}, r^{VIX})$	$\rho(r^{VSR}, r^{str})$
Australian Dollar	-2.08	-0.14	-6.37	-0.41	0.05	0.00	0.99	0.41
British Pound	-9.40	-0.46	-11.82	-0.58	-0.51	-0.03	0.96	0.31
Canadian Dollar	-8.81	-0.52	-13.46	-0.82	-3.91	-0.21	0.99	0.35
Swiss Franc	14.79	0.19	14.20	0.17	6.44	0.27	1.00	0.39
Japanese Yen	-16.56	-0.98	-19.55	-1.20	-3.93	-0.17	0.97	0.42
S&P 500	-11.93	-0.23	-13.08	-0.24	-0.75	-0.04	1.00	0.51
10-year UST	-25.79	-2.97	-31.94	-3.68	-6.14	-0.29	0.97	0.53
30-year UST	-14.44	-1.41	-18.77	-1.76	5.18	0.21	0.99	0.60
Brent Crude Oil	-12.17	-0.46	-17.18	-0.61	-8.98	-0.59	0.99	0.30
Natural Gas	-14.13	-0.80	-15.31	-0.86	-13.45	-0.68	1.00	0.44
Gold	-11.71	-0.42	-12.70	-0.47	7.45	0.30	1.00	0.41
Silver	-5.50	-0.11	-3.69	-0.07	6.43	0.32	0.89	0.49
Copper	-17.64	-1.41	-18.73	-1.35	-13.30	-0.65	0.92	0.45
Live Cattle	-21.95	-1.52	-25.17	-1.77	-5.93	-0.27	1.00	0.60
Soybean Meal	-15.66	-1.40	-23.22	-2.07	-0.65	-0.03	0.99	0.16
Soybean Oil	-16.69	-1.40	-19.15	-1.63	-10.00	-0.42	0.95	0.17
Soybeans	-22.91	-1.95	-24.51	-2.14	4.82	0.20	0.99	0.35
Wheat	-13.79	-1.30	-15.16	-1.41	-5.83	-0.32	0.99	0.18
Coffee C	-18.49	-1.57	-23.26	-2.00	-7.23	-0.34	0.98	0.54
Corn	-23.26	-1.71	-22.79	-1.67	-8.51	-0.37	0.76	0.50

Fig. A.1. VSR and hedged VIX^2 returns. The graphs show variance swap returns (in blue) and hedged VIX^2 returns (in black) for Australian dollar, British pound, S&P 500, 10-year Treasuries, gold and natural gas.

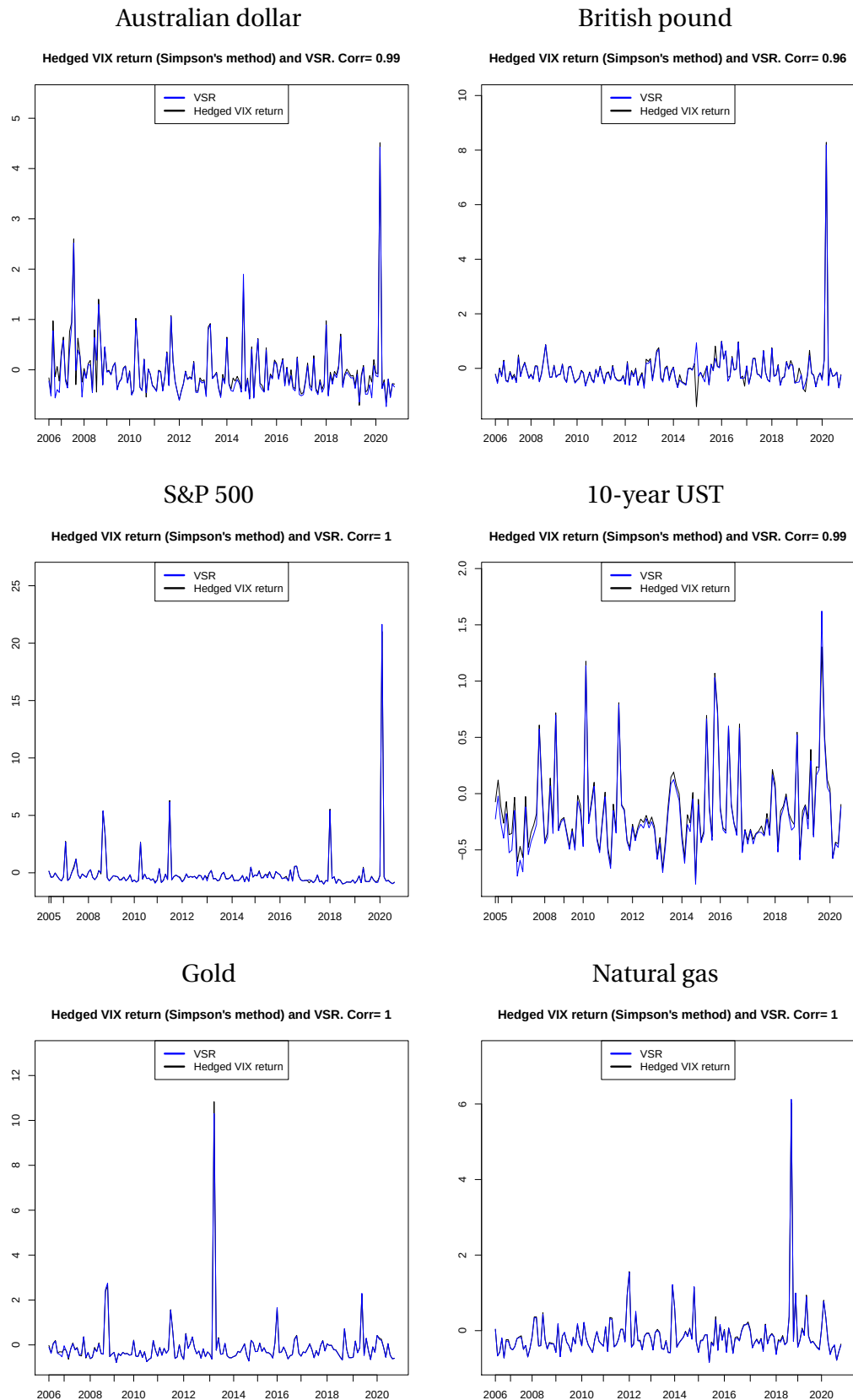
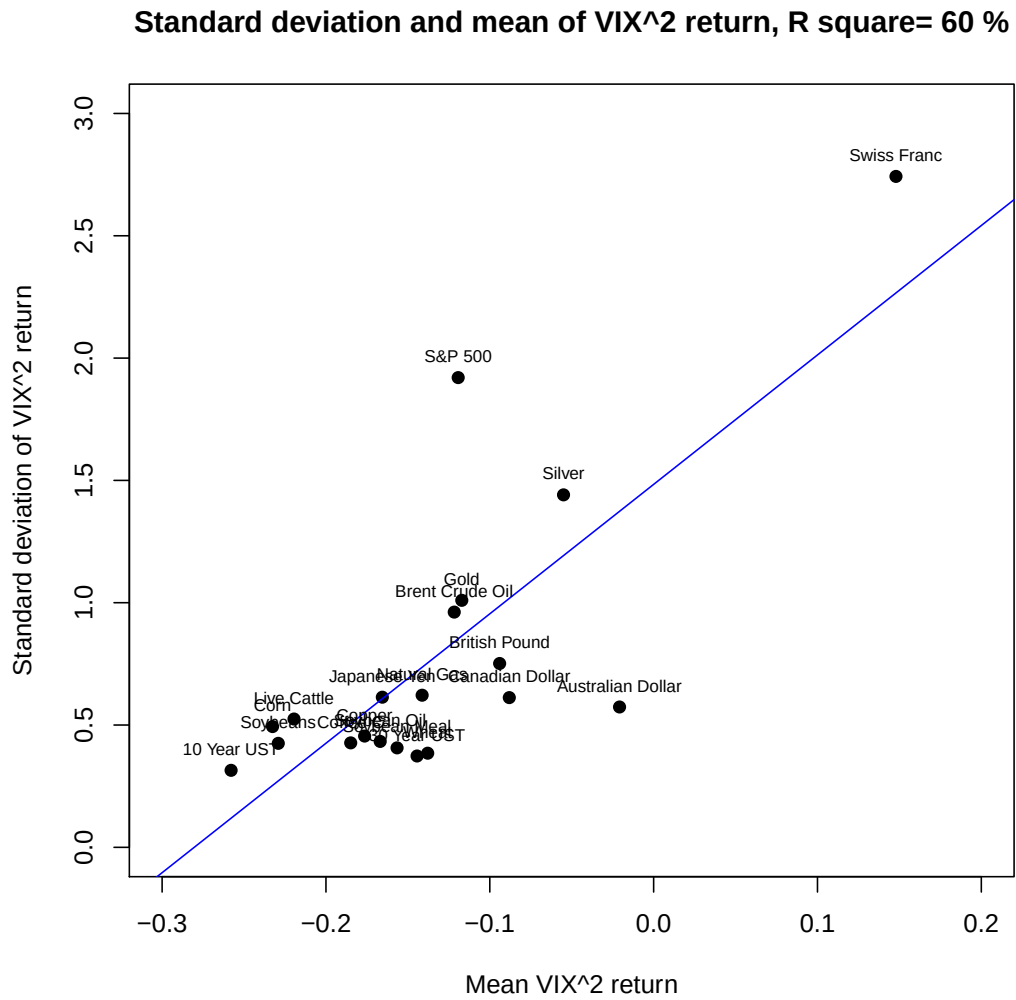


Fig. A.2. Scatterplot of the mean VIX^2 return and its standard deviation. The figure shows the average VIX^2 return over time and its standard deviation.



Appendix B: Some details about the data and robustness checks

American vs European. Most of the options we analyze are American, while the VIX CBOE formula assumes European options. For robustness, we recalculated the VIX^2 returns using European option prices (based on the implied volatilities of the American ones) and found that the results are nearly identical. The fact that using American options instead of European ones does not have a substantial impact on the variance risk premium returns is consistent with other studies: e.g., [Choi et al. \(2017\)](#), [Dew-Becker et al. \(2017\)](#).

Convenience yields. The convenience yield of futures cancels out in the dynamic hedge term and the static position. Hence, the existence of convenience yields does not change the results.

Futures vs forwards. We expect the difference due to using futures vs forwards to be small. For example, in the Eurodollar futures market, [Flesaker \(1993\)](#) and [Cakici and Zhu \(2001\)](#) show that the impact of using futures as the underlying as opposed to forwards is very small particularly for options with short maturity.

Details about the Beason and Schreindorfer calculation. We estimate average payoffs, $f(R) = \mathbb{E}[1\{R < R_{t:t+30} < R + dR\}]$, with the empirical PDE, which attaches a probability of $1/N$ to each return observation for the N days in our sample. By sampling daily, we are able to estimate f based on 21 times more observations than would be the case under monthly sampling, which yields substantially more precision, just like in [Beason and Schreindorfer \(2022\)](#).

To obtain $f^*(R)$, we first calculate implied volatilities (IVs) from observed option prices across strikes on a given day using Black's formula (a version of [Black and Scholes \(1973\)](#)). We then interpolate the implied volatility surface across strikes and maturities using the Stochastic Volatility Inspired (SVI) model, which is widely used in financial institutions due to its parsimonious nature and excellent fit to observed IVs. The SVI model describes implied variance (the square of IV) for a given maturity T with the function:

$$\sigma_{\text{BSM}}^2(x) = a + b(\rho(x - m) + \sqrt{(x - m)^2 + \omega^2}),$$

where $x = \log(K/F_{t,T})$ is the option's log-moneyness, $F_{t,T}$ is the forward price for maturity T , and a, b, ρ, m, ω are parameters. We fit the model to the 1 month maturity.

For each day in the sample, we estimate the SVI parameter vector $\theta = (a, b, \rho, m, \omega)$ by minimizing the root mean squared error between observed IVs and the SVI interpolant:

$$\hat{\theta}_t = \arg \min_{\theta} \frac{1}{N_t} \sum_{i=1}^{N_t} [\sigma_{\text{BSM},t,i} - \sigma_{\text{BSM}}(\kappa_{t,i}, T_{t,i}; \theta)]^2,$$

where N_t is the number of observations on day t . We discard parameters for which SVI-implied prices violate no arbitrage constraint and check the positivity, monotonicity, and convexity of option prices.

After having all implied volatilities, we calculate the corresponding call option prices and compute f_t^* using [Breedon and Litzenberger \(1978\)](#). Specifically, the absence of arbitrage opportunities implies that the time- t price of a European call option with strike K and maturity $t + T$ equals

$$C_t(K, T) = \frac{1}{R_t^f} \int_K^\infty (S_{t+T} - K) f_t^*(S_{t+T}) dS_{t+T},$$

the risk-neutral expectation of its payoff discounted at the risk-free rate. Differentiating this expression twice with respect to K yields the conditional risk-neutral PDF of the future price level as

$$f_t^*(S_{t+T}) = R_t^f \frac{\partial^2 C_t(K, T)}{\partial K^2} \Big|_{K=S_{t+T}}.$$

The risk-neutral PDF of ex-dividend returns follows from the change of variables for the 1-month return $R_{t:t+30} = \frac{S_{t+30}}{S_t} - 1$ as

$$f_t^*(R_{t:t+30}) = S_t f_t^*(S_{t+30}).$$

We integrate f_t^* over the κ -region and discard parameters for which these integrals do not fall within (a numerical error tolerance of) 1 basis point of one. We then finally obtain $f^*(R) = \frac{1}{N} \sum_{t=1}^N f_t^*(R)$, where N is the number of days in our sample.